

Executive Summary

This project analyzes the shipping efficiency of Nassau Candy Distributor by evaluating factory-to-customer deliveries across different regions, states, and shipping modes. Using Power BI, Power Query, and DAX, key metrics such as Total Sales, Total Profit, Total Orders, Profit Margin, and Average Lead Time were analyzed. The dashboard helps identify delivery bottlenecks, compare shipping performance, and evaluate product profitability. Based on the insights, recommendations were provided to improve logistics efficiency, optimize shipping operations, and support data-driven business decisions.

Tools & Technologies Used

- **Power BI** – Dashboard Development & Data Visualization
- **Power Query** – Data Cleaning & Transformation
- **DAX** – KPI & Measure Creation
- **Microsoft Excel** – Data Review & Validation
- **Business Analytics Techniques** – KPI Analysis, Performance Measurement, and Recommendation Development

Business Problem

Nassau Candy Distributor lacks clear visibility into the efficiency of its factory-to-customer shipping operations. The organization needs to identify which regions and shipping methods perform efficiently, where delivery delays occur, and how shipping performance impacts overall business operations. Without these insights, logistics decisions remain reactive rather than data-driven, leading to potential delays, increased operational costs, and reduced customer satisfaction.

Key Questions Addressed:

- Which regions experience the highest shipping lead times?
- Which shipping modes are most efficient?
- Where do delivery bottlenecks exist?
- Which products and divisions contribute most to sales and profitability?
- How can shipping and operational efficiency be improved?

Dataset Overview

The dataset contains order, shipment, customer, product, and profitability information for Nassau Candy Distributor. It includes details related to sales transactions, shipping dates, delivery locations, product categories, and shipping methods. The data was used to analyze shipping efficiency, identify delivery bottlenecks, and evaluate product and regional performance.

Key Fields Used

Category	Fields
Order Information	Order ID, Order Date, Ship Date
Geography	City, State/Province, Region
Product Information	Product Name, Division
Shipping	Ship Mode
Sales Metrics	Sales, Units, Gross Profit
Calculated Metrics	Lead Time Days, Profit Margin %

🔍 Search

▼ 📊 Nassau Candy Distributor ...

☐ 📊 Avg Lead Time Test

☐ City

☐ Σ Cost

☐ Country/Region

☐ Σ Customer ID

☐ Division

☐ Σ Gross Profit

☐ Σ Lead Time Days

> ☐ 📅 Order Date

☐ Order ID

☐ Postal Code

☐ Product ID

☐ Product Name

☐ 📊 Profit Margin %

☐ Region

☐ Σ Row ID

☐ Σ Sales

> ☐ 📅 Ship Date

Dataset Statistics

- Total Orders: **95**
- Regions Covered: **4**
- Ship Modes: **4**
- Product Divisions: **4**
- Analysis Period: **2024–2027**

Data Cleaning & Validation

To ensure accurate analysis, the dataset was cleaned and validated using Power Query. Date fields were converted to the appropriate date format, and a new **Lead Time Days** column was created by calculating the difference between Order Date and Ship Date. The dataset was checked for missing shipment records, duplicate entries, and invalid lead times. Geographic fields such as Region and State/Province were standardized to maintain consistency across visualizations. These steps improved data quality and ensured reliable insights for shipping efficiency and profitability analysis.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID
1	US-2021-103809-CHO-MB-32900	1/3/2024	4/30/2026	Standard Class	
2	US-2021-112026-CHO-TR-04000	1/4/2024	7/1/2026	Standard Class	
3	US-2021-112326-CHO-NUT-13000	1/4/2024	7/1/2026	Standard Class	
4	US-2021-112326-CHO-SCB-58200	1/4/2024	7/1/2026	Standard Class	
5	US-2021-141817-CHO-TR-04000	1/5/2024	7/5/2026	Standard Class	
6	US-2021-167199-CHO-SCB-58200	1/6/2024	7/5/2026	Standard Class	
7	US-2021-167199-CHO-TR-04000	1/6/2024	7/5/2026	Standard Class	
8	US-2021-166254-CHO-MB-32900	1/6/2024	4/30/2026	First Class	
9	US-2021-167199-CHO-NUT-13000	1/6/2024	7/5/2026	Standard Class	
10	US-2021-167199-CHO-MB-32900	1/6/2024	7/5/2026	Standard Class	
11	US-2021-167199-CHO-MB-32900	1/6/2024	7/5/2026	Standard Class	
12	US-2021-130813-CHO-MB-32900	1/6/2024	7/5/2026	Standard Class	
13	US-2021-167199-CHO-SCB-58200	1/6/2024	7/5/2026	Standard Class	
14	US-2021-167199-CHO-FUD-93000	1/6/2024	7/5/2026	Standard Class	
15	US-2021-105417-CHO-MB-32900	1/7/2024	7/5/2026	Standard Class	
16	US-2021-105417-CHO-TR-04000	1/7/2024	7/5/2026	Standard Class	
17	US-2021-135405-CHO-TR-04000	1/9/2024	7/6/2026	Standard Class	
18	US-2021-135405-CHO-TR-04000	1/9/2024	7/6/2026	Standard Class	
19	US-2021-149030-CHO-MB-32900	1/10/2024	7/8/2026	Standard Class	
20	US-2021-149030-CHO-GUNP-210...	1/10/2024	7/8/2026	Standard Class	

KPI Definitions

KPI	Description
Total Sales	Total revenue generated from all orders
Total Profit	Total gross profit earned
Total Orders	Number of unique customer orders
Total Units Sold	Total quantity of products sold
Average Lead Time	Average number of days between order and shipment
Profit Margin %	Profit generated as a percentage of sales

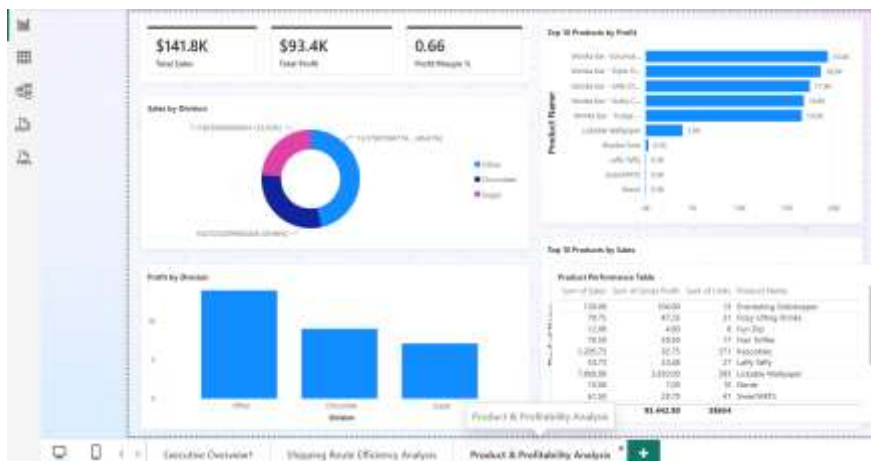
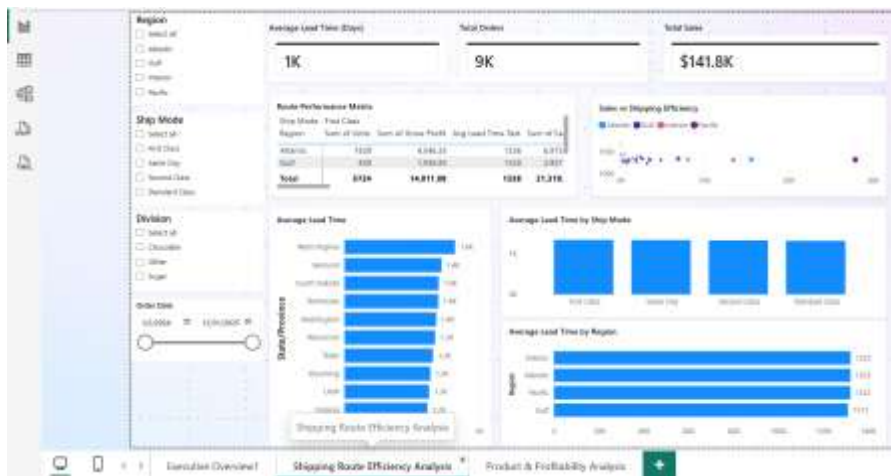
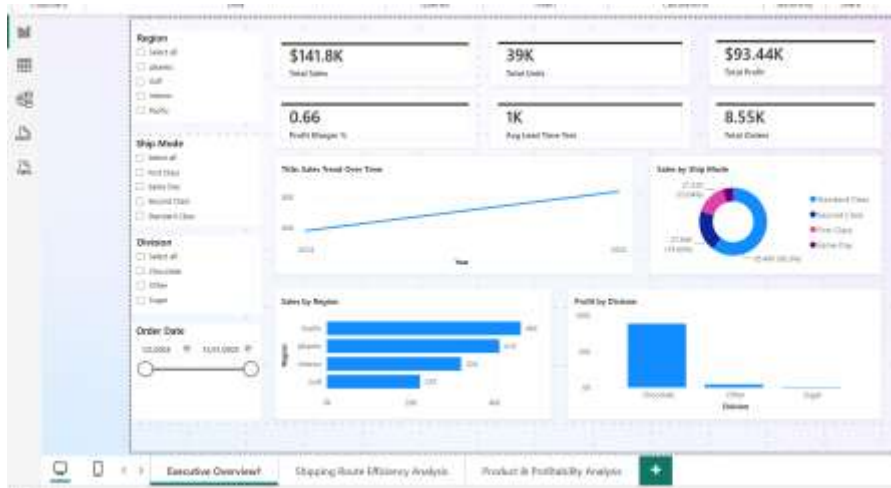
Dashboard Insights & Analysis

The dashboard analysis highlights key trends in shipping performance, sales, and profitability across different regions and product categories. Regional analysis revealed variations in average lead times, indicating potential delivery bottlenecks in specific locations. Shipping mode comparison showed differences in delivery performance and operational efficiency. Product analysis identified top-performing products and divisions contributing the highest sales and profit. The profitability analysis further highlighted opportunities to focus on high-margin products and optimize business performance. Overall, the dashboard provides a comprehensive view of operational efficiency and supports data-driven decision-making.

Key Findings

- The Interior region generated the highest sales contribution among all regions.
- Shipping lead times varied across regions and states, indicating opportunities for logistics optimization.
- Certain shipping modes demonstrated better delivery performance than others.
- A small number of products contributed a significant share of overall sales and profit.

- Profitability was concentrated within specific product divisions, highlighting key revenue drivers.
- Regional performance differences suggest the need for targeted operational improvements.



Recommendations

Based on the analysis, the following recommendations are proposed to improve shipping efficiency and business performance:

- Optimize logistics operations in regions and states with higher average lead times.
- Monitor and improve the performance of shipping modes associated with delivery delays.
- Prioritize high-performing products and divisions that contribute significantly to sales and profitability.
- Implement data-driven inventory planning to ensure product availability for high-demand items.
- Regularly track shipping and profitability KPIs through dashboards to support proactive decision-making.
- Conduct further investigation into regional bottlenecks to reduce delivery delays and improve customer satisfaction.

Conclusion

This project successfully analyzed factory-to-customer shipping efficiency using Power BI and business intelligence techniques. Through data cleaning, KPI development, and interactive dashboard creation, valuable insights were generated regarding sales performance, shipping operations, and product profitability. The findings help identify operational bottlenecks and support data-driven decision-making. By implementing the recommended actions, Nassau Candy Distributor can improve logistics efficiency, enhance customer service, and drive overall business performance.